

AE 2610, Lab 3

Dynamics

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Lab Group A02-X

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Introduction

The purpose of this lab was to measure and understand the response of a dynamical system over time. More specifically, this lab revolved around using raw data to estimate and find material characteristics, like spring constants or damping coefficients, as well as deriving equations to describe the motion of a system. This lab consisted of four main configurations (linear vs. non-linear, zero blocks vs. four blocks) as well as a supplementary configuration.

In this lab, the one-dimensional displacement of a cart was measured utilizing a 1-DOF sliding cart system with an embedded linear optical shaft encoder. This optical encoder provided information on the instantaneous movement of the cart by sensing alternating intensity of light, using LEDs and photodetectors. When connected to a data acquisition device (computer), it became possible to use this optical information to map the displacement of the cart in real-time. It is important to note that the 1-DOF sliding cart mechanism was oriented 45° from the edge of the table to minimize the external forces. Additionally, the mass of all the individual components that make up a configuration were measured using a digital scale. To measure the weight of the cart, it was essential to hang the cart vertically to ensure that the mass of the greater sliding mechanism was not measured too. With this combination of raw data, it becomes possible to estimate material characteristics and identify the governing equations for the cart's motion.

Data Results

Raw Data

Human Measurements

Lab members were responsible for measuring the mass of all the different individual components that made up the cart in both the zero blocks and four blocks configurations. More specifically, each individual component was measured using a standard digital scale, recording mass up to two decimal points. To measure the weight of the cart, it was essential to hang the cart vertically to ensure that its mass was isolated and that the mass of the greater sliding mechanism was not included. Additionally, the mass of small components, like screws/washers and nuts/bolts, were taken together to simplify further calculations. Once all the individual components were measured, the total mass of each configuration needed to be noted by simply summing the mass of the components it included. The human measurements taken in this lab are best seen in Table I.

Computer Measurements

In addition to the measurements made by the lab members, computer measurements were acquired through the use of linear optical shaft encoders attached to the sliding mechanism of the cart. More specifically, the 1 degree-of-freedom (1-DOF) sliding cart mechanism can translate along a low-friction ball slider while containing an encoder and encoder slide. It is important to note that the 1-DOF sliding cart mechanism was oriented 45 degrees from the edge of the table in order to minimize the impact on any accidental bumps. The linear optical encoder embedded in the sliding cart mechanism provides information on the instantaneous movement of the cart by combining LEDs and photodetectors to sense alternating intensity of light and convert the results to displacement. With this setup, the computer was able to generate thousands of raw data points

in the form of five MATLAB files. (.mat) There was one file for each different configuration, including the supplement nonlinear configuration. All in all, the total accumulation and representation of the raw data points for each configuration can best be seen with Figure 1 through Figure 5.

Reduced Data

Time Window of Interest Identification

The first step in the data reduction of this lab was to find the specific time windows of interest for each different configuration. The overall process of determining these time windows was not too difficult. Simply examine the raw data, identify the point where the cart was released as the start point, and identify the point where the oscillations have settled as the end point. It is essential to first isolate the time windows of interest before attempting to find material properties, like spring constants and damping coefficients, because the raw data before and after this window will significantly skew the results. Overall, the time windows of interest for each configuration can be found in Table III.

Linear System Response Spring Constant and Damping Coefficient

With correct time windows of interest, it is almost possible to start thoroughly analyzing the system responses of the two linear configurations. Before this analysis can begin, the spring constant of Spring 1, k_1 , and the damping coefficient, b , must be estimated.

First, an accurate estimate of the spring constant of Spring 1 can be made by utilizing simple harmonic motion. With oscillations, the time period of movement can be represented in equation (1) below:

$$T = 2\pi \sqrt{\frac{m}{k_1}} \quad (1)$$

Using this knowledge, simply rewrite equation (1) to solve for k_1 as seen below:

$$k_1 = m \left(\frac{2\pi}{T} \right)^2 \quad (2)$$

To solve for k_1 , simply choose a linear configuration, measure the time of one complete oscillation, and insert the values into equation. Using the linear zero block configuration, for example, the mass of the configuration was 0.66011 kg, and the period was 0.552 seconds, giving a spring constant estimation of 85.5 N/m.

Similarly, the damping coefficient, b , can also be estimated by looking at how an oscillation changes over time when naturally damped (Friction/Air Resistance). The displacement of the 1-D damped oscillation can be represented in equation (3) below:

$$x(t) = A_0 e^{-\zeta \omega_n t} \cos(\omega_d t + \phi) \quad (3)$$

In order to simplify this expression and solve for the damping coefficient, it is possible to get ratio between the initial peak x_1 and the subsequent peak x_2 that occurs one oscillation later.

This ratio is best expressed in equation (4) below:

$$\frac{x_1}{x_2} = \frac{e^{-\zeta \omega_n t_1}}{e^{-\zeta \omega_n (t_1 + T)}} = e^{\zeta \omega_n T} \quad (4)$$

Continuing on, this expression can be further simplified by removing the exponent all together. This simplification is possible by taking the natural log of both sides as seen in equation (5) below:

$$\ln \left(\frac{x_1}{x_2} \right) = \zeta \omega_n T \quad (5)$$

Additionally, it is extremely important to note that the damping ratio can be expressed in the following form:

$$\zeta = \frac{b}{2m\omega_n} \quad (6)$$

With this understanding, it becomes possible to simplify again and remove both the damping ratio and natural frequency from the equation altogether. This is best illustrated in equation (7) as seen below:

$$\ln\left(\frac{x_1}{x_2}\right) = \left(\frac{b}{2m\omega_n}\right) \omega_n T \quad (7)$$

Finally, in order to achieve an accurate estimate for the value of the damping coefficient, b , rewrite equation (8) to solve for it.

$$b = \frac{2m}{T} \ln\left(\frac{x_1}{x_2}\right) \quad (8)$$

Utilizing equation (8), simply chose either linear configuration to solve for an estimate of the damping coefficient. With the linear zero block configuration, simply insert the mass (0.66011 kg), period (0.552 s), and heights of the first two oscillations ($x_1 = 0.0479901$, $x_2 = 0.0453549$) to get the final damping coefficient estimate of 0.1351 Ns/m.

Trial and error was used to find the actual values of the spring constant for Spring 1 as well as the damping coefficient. Utilizing MATLAB, simply insert the linear equation of motion (equation (17/18)), and alter the two variables until the raw data and linear model curves overlap. Using this methodology, it was found that the actual Spring 1 constant was 81.77 N/m, and the actual damping coefficient was 0.1668 Ns/m. All in all, this data can be found summarized in Table IV.

Linear System Response Damping Ratio and Natural Frequency

In order to find the damping ratio and natural frequency in the linear configurations, the following two equations must be rewritten:

$$\ddot{x}(t) + \frac{b}{m} \dot{x}(t) + \frac{k}{m} x(t) = \frac{F_{external}(t)}{m} \quad (9)$$

$$\ddot{x}(t) + 2\zeta\omega_n \dot{x}(t) + \omega_n^2 x(t) = \frac{F_{external}(t)}{m} \quad (10)$$

First, set up a ratio comparing the coefficients of $x(t)$ from both equations as seen in equation (11) below:

$$\omega_n^2 = \frac{k_1}{m} \quad (11)$$

Here, the natural frequency can easily be calculated by taking the square root of both sides as seen below:

$$\omega_n = \sqrt{\frac{k_1}{m}} \quad (12)$$

Second, set up a ratio comparing the coefficients of $x'(t)$ from both equations as seen in equation (13) below:

$$2\zeta\omega_n = \frac{b}{m} \quad (13)$$

In order to solve for the damping ratio, make sure to substitute in the natural frequency using the logic of equation (14) as seen below:

$$2\zeta\sqrt{\frac{k_1}{m}} = \frac{b}{m} \quad (14)$$

Now, all that is left is to rearrange the expression to solve for the damping ratio. This logic and rearrangement is expressed in the equation below:

$$\zeta = \frac{b}{2m\sqrt{\frac{k_1}{m}}} = \frac{b}{2\sqrt{k_1 m}} \quad (15)$$

Once the natural frequency and damping ratio are arranged in terms of the model parameters, it becomes possible to solve by simply inserting the corresponding mass, spring constant, and damping coefficient. For the linear zero block configuration, the natural frequency is 11.13 rad/s and the damping ratio is 0.015. For the linear four block configuration, the natural frequency is 6.78 rad/s and the damping ratio is 0.019.

Linear Equation of Motion Derivation

Compared to the non-linear equation of motion, the linear equation of motion is relatively simple and can be expressed in the following equation:

$$m\ddot{x} + b\dot{x} + k_1x = 0 \quad (16)$$

In order to find the real damping coefficient and spring constant values, simply use trial and error until the linear model overlaps with the raw data. The final values for the damping coefficient and spring constant can be found in Table IV. Simply insert the different masses of each configuration to get their respective linear equation of motion.

Zero Blocks Configuration:

$$0.66011\ddot{x} + 0.1688\dot{x} + 81.77x = 0 \quad (17)$$

Four Blocks Configuration:

$$1.85348\ddot{x} + 0.2268\dot{x} + 81.77x = 0 \quad (18)$$

Distance Between Cart and Free-Hanging Spring (Spring 4) Analysis

In order to start analyzing the non-linear configurations of this lab, it is necessary to know the separation distance between the cart and the damper (Spring 4). The main approach for finding this critical distance was to analyze the difference in the time it took the cart to hit its peak position between successive peaks. By analyzing the change in time it took the cart to hit successive peaks, it was clear that a plateau occurred around the 11th peak of the nonlinear zero blocks configuration. To find the distance between the cart and the free-hanging spring (Spring 4), simply identify the y-value (displacement) at this peak on Figure 3. This distance is expressed in Table V, turning out to be around 0.0244 m or 24.4 mm.

Non-Linear System Response Second Spring Constant

Estimating the spring constant of Spring 4 in the non-linear configurations followed a similar logic to estimating the spring constant of Spring 1 in the linear configurations. With this being said, there are two major differences in the equation (19) that can be expressed below:

$$k_{eff} = m \left(\frac{2\pi}{T_{nonlinear}} \right)^2 \quad (19)$$

First, it is very important to understand that Spring 1 and Spring 4 are acting in series when the nonlinear configuration is being actively dampened. This means that the spring constants of both Spring 1 and Spring 2 must be combined together to form the effective spring constant of the system. Since the two springs are in series, this relationship can be expressed in the equation below:

$$k_{eff} = k_1 + k_2 \quad (20)$$

Second, it is very important to understand that Spring 4 does not act on the cart for the entire period of oscillation. In fact, Spring 4 can only dampen the cart system when it is past the separating distance of 24.4 mm in the positive direction of displacement. This unique impact is the main reason as to why as the positive displacement of the nonlinear configurations is considerably shorter than the negative displacement as seen in Figure 3 and Figure 4. With this shortened period in mind, the following equation (21) can be used to accurately estimate the value of the Spring 4 constant:

$$k_2 = m \left(\frac{2\pi}{T_{nonlinear}} \right)^2 - k_1 \quad (21)$$

With equation (21), using the spring constant of the damper (Spring 4) can be found to be 289.4 N/m by using the mass (1.85348 kg) and shortened period (0.22 s) of the non-linear four blocks configuration as well as the spring constant of Spring 1 (81.77 N/m).

Trial and error was used to find the real values of the spring constant for Spring 4. Utilizing MATLAB, simply insert the nonlinear equation of motion (equation (25/26)), and alter

the k_2 and b_2 variables until the raw data and nonlinear model curves overlap. Using this methodology, it was found that the actual Spring 4 constant was 253.0 N/m. All in all, this data can be found summarized in Table IV.

Non-Linear Heaviside Function Derivation

In order to describe the motion of the nonlinear configurations, a Heaviside function must be implemented. A Heaviside function is a step function that is zero for negative arguments and one for positive arguments. This specific type of function is essential for describing the impact of the Spring 4 damper. As mentioned earlier, the damper only impacts the motion of the cart when in contact. In other words, the damper only contributes to the motion equation when the cart has a displacement over 24.4 mm. Expressed as a piecewise function, the impact of Spring 4 can be represented below:

$$F_{spring4} = \begin{cases} 0, & \text{if } x \leq 0.0244 \\ k_2(x - 0.0244), & \text{if } x > 0.0244 \end{cases} \quad (22)$$

By converting this piecewise function into a Heaviside function, equation (23) better represents this relation below:

$$F_{spring2} = k_2(x - x_c) \cdot H(x - x_c) \quad (23)$$

Finally, through trial and error, simply insert the damping coefficient, both spring constants, and each non-linear configuration's mass into the following equation:

$$m\ddot{x} + b\dot{x} + k_1x + k_2(x - 0.0224)H(x - 0.0224) = 0 \quad (24)$$

Zero Blocks Configuration:

$$0.66011\ddot{x} + 0.1668\dot{x} + 87.11x + 253(x - 0.0224)H(x - 0.0224) = 0 \quad (25)$$

Four Blocks Configuration:

$$1.85348\ddot{x} + 0.2268\dot{x} + 87.11x + 253(x - 0.0224)H(x - 0.0224) = 0 \quad (26)$$

Effective Spring Stiffness Expression

Understanding the logic behind the Heaviside function, it is important to note that the effective spring stiffness of a nonlinear configuration can also be represented as a piecewise function. It is important to note that the effective spring stiffness cannot be represented using a Heaviside function, because there is always at least one spring acting on the cart. With this being said, the effective spring stiffness can be represented in the expression below:

$$k_{eff}(x) = \begin{cases} k_1, & \text{if } x \leq 0.0244 \\ k_1 + k_2, & \text{if } x > 0.0244 \end{cases} \quad (27)$$

Here, the effective spring constant switches from k_1 to $k_1 + k_2$ depending on whether the cart is in contact with the damper (Spring 4). With this being said, the actual effective spring constant in the non-linear configurations can be represented below:

$$k_{eff}(x) = \begin{cases} 81.77 \text{ N/m}, & \text{if } x \leq 0.0244 \\ 334.79 \text{ N/m}, & \text{if } x > 0.0244 \end{cases} \quad (28)$$

Discussion

Supplement Questions

1. How well do the spring constant values match for both mass setups?

The spring constant values for Spring 4 in both mass setups were almost identical. These two values only varied by less than 10 N/m, and this small discrepancy is most likely due to human error when moving the cart or contact error when the cart hit Spring 4.

2. What are the differences in damping coefficients for the different mass setups and why they may exist?

The damping coefficient for the four blocks configuration was consistently higher than the damping configuration of the zero blocks configuration. This discrepancy exists because additionally weight leads to an increase of friction on the chart. Since this force is unaccounted, the damping coefficient of high weight configurations will tend to be higher.

3. What are the differences in the numerical vs. experimental plots for nonlinear response?

The main differences between the nonlinear numerical vs. experimental plots arise when the damper is no longer in contact with the cart. The numerical plot assumes that this transition is instantaneous, whereas the experimental plot is more gradual. Additionally, unaccounted forces, like friction and air resistance, create significant differences towards the end of oscillation.

4. What are the differences in analytical vs. experimental plots for linear response?

The main differences between the linear analytical vs. experimental plots are found towards the end of the configuration's oscillation. This discrepancy is due to unaccounted forces, like friction and air resistance, prematurely slowing down the experimental plot over time.

5. Why is this system non-linear?

Spring 4 was the reason why the zero blocks and four blocks configurations became non-linear. Spring 4 acted as a damper, restricting the movement of the cart and disproportionately absorbing kinetic energy when the cart passed the offset distance.

6. How was the offset distance of the second spring determined in the non-linear experiment?

The offset distance of the second spring/damper was determined in the non-linear experiment by analyzing the change in time it took the cart to hit successive peaks, finding where these peaks plateaued, and identifying the overall displacement at this peak.

7. The supplemental non-linear responses. Are they actually non-linear? If so, what makes the response non-linear? When would you see something like this?

The supplement non-linear response was the most non-linear configuration. This configuration was the most non-linear because Spring 4 still acted as a damper with an additional ruler damper acting in the negative direction. This dual damper configuration is often used in cars, where one ‘soft’ spring is used for standard operation before converting to another stiff spring during extreme impact.

Additional Discussion

- The supplemental non-linear portion of this lab involved the addition of a ruler to act as a second damper in addition to Spring 4.
- The non-linear model in Figure 10 is considerably different than the raw data points because the application of the ruler was both variable and unmeasured.

Tables and Figures

Table I. Mass measurements and configurations.

Configuration	Zero Weight Blocks	Four Weight Blocks
Cart Mass (g)	560.64	560.64
Electromagnet Mass (g)	32.60	32.60
Screw & Washer Mass (g)	4.54	4.54
Interface Block Mass (g)	62.33	124.66
Weight Block Mass (g)	0	1110.44
Nut & Bolt Mass (g)	0	20.60
Total Mass (g)	660.11	1853.48

Table II. Spring Selection

Configuration	Zero Weight Blocks	Four Weight Blocks
Linear	Spring 1	Spring 1
Nonlinear	Spring 1 & Spring 4	Spring 1 & Spring 4

Table III. Time windows of interest for each configuration

Configuration	Start Time (s)	End Time (s)
Linear Zero Blocks	2.974	15.564
Linear Four Blocks	10.387	24.516
Nonlinear Zero Blocks	4.432	14.914
Nonlinear Four Blocks	13.384	26.037

Supplemental Nonlinear	4.978	11.255
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Table IV. Linear response initial guesses for spring constant and damping coefficient.

Parameter	Spring Constant k_1 (N/m)	Damping Coefficient b (Ns/m)
Initial Guess	85.5	0.1351
Final Adjusted Measure	81.77	0.1668

Table V. Non-linear response initial guesses for offset distance and second spring constant.

Parameter	Offset Distance (m)	Spring Constant k_2 (N/m)
Initial Guess	X	289.4
Final Adjusted Measure	0.0244	253.0

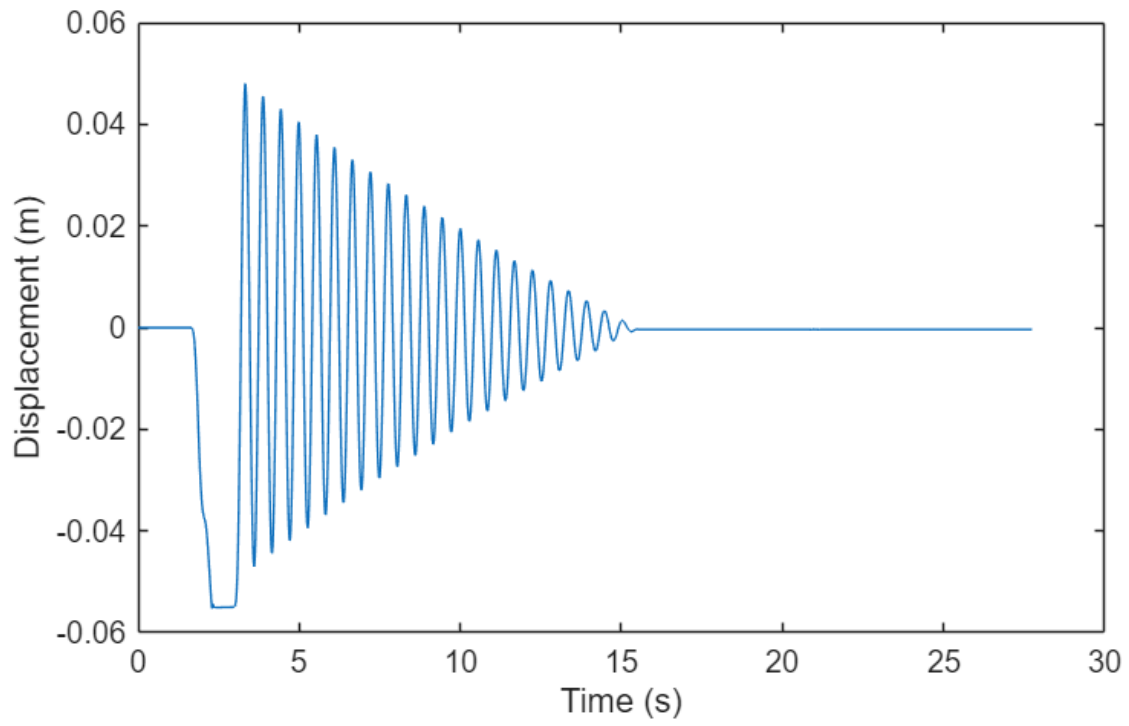


Figure 1. Raw data for linear response of zero block configuration.

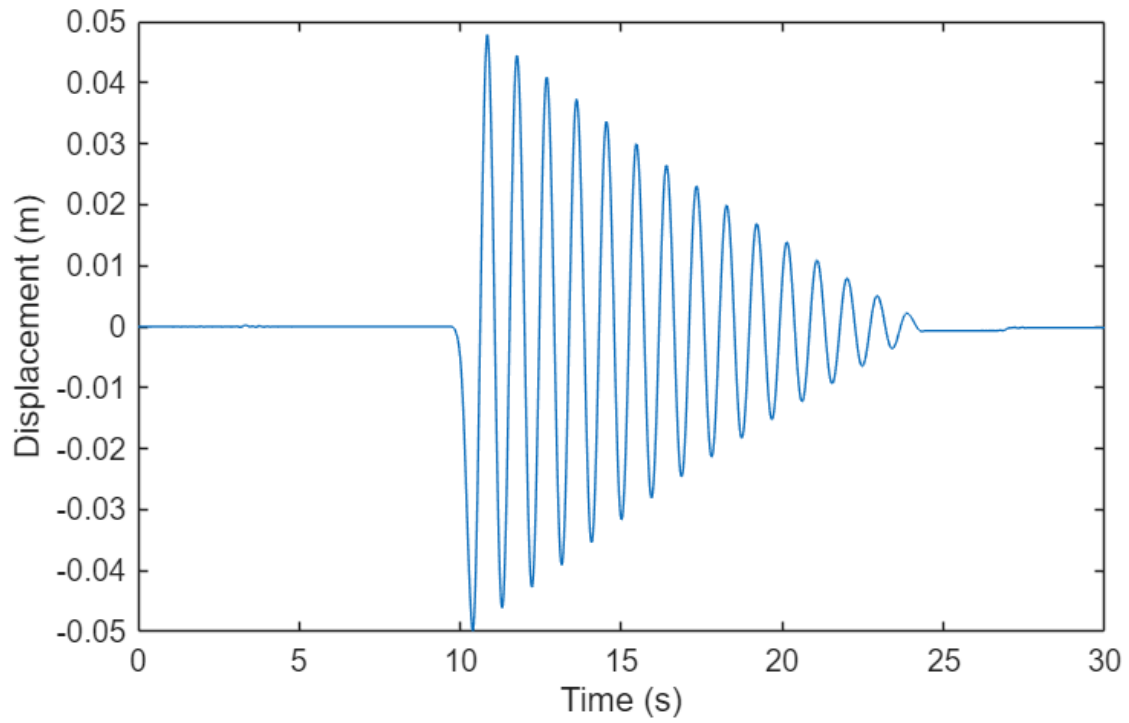


Figure 2. Raw data for linear response of four blocks configuration.

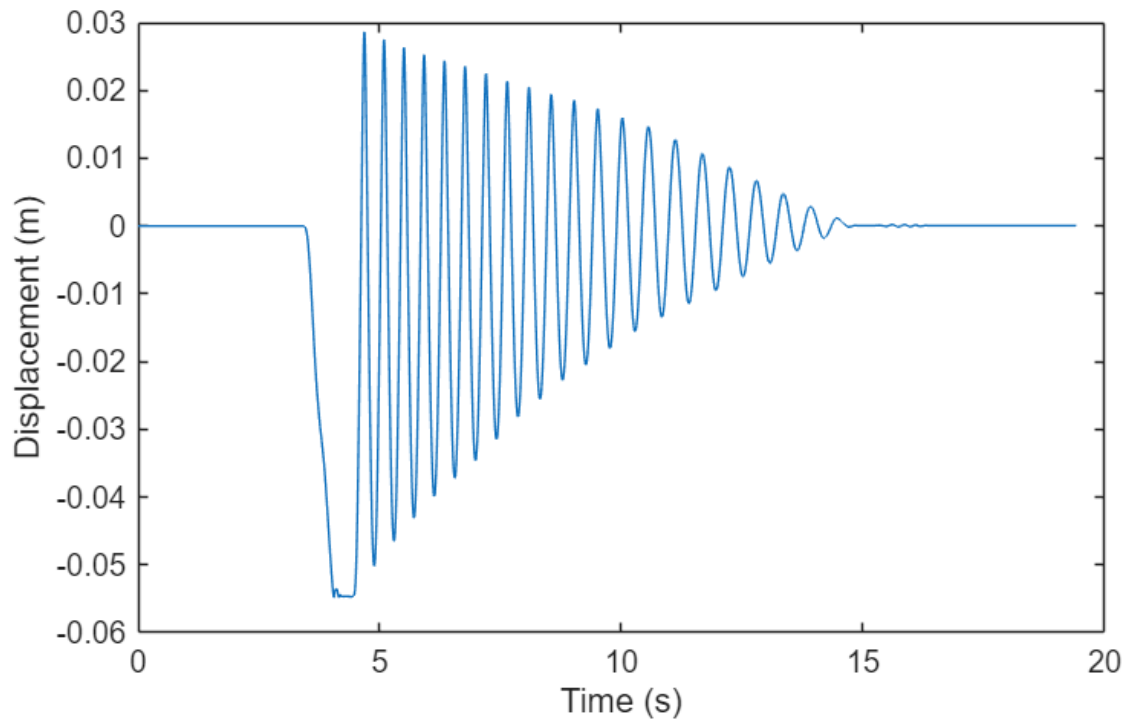


Figure 3. Raw data for non-linear response of zero block configuration with.

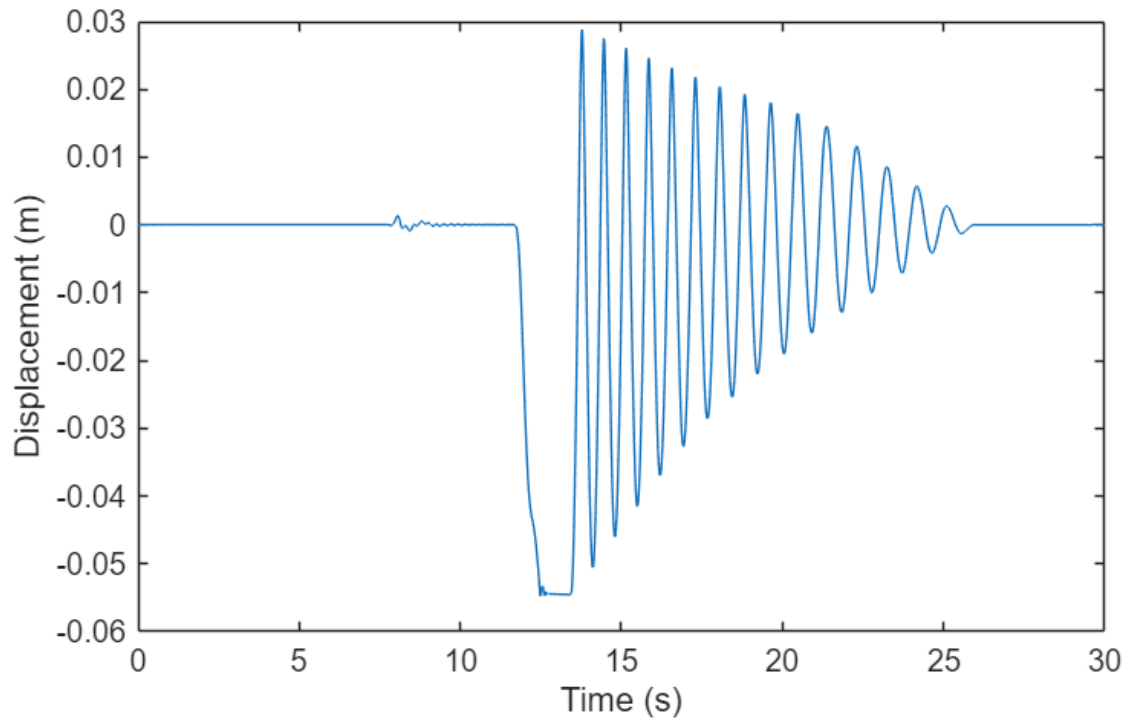


Figure 4. Raw data for non-linear response of four blocks configuration.

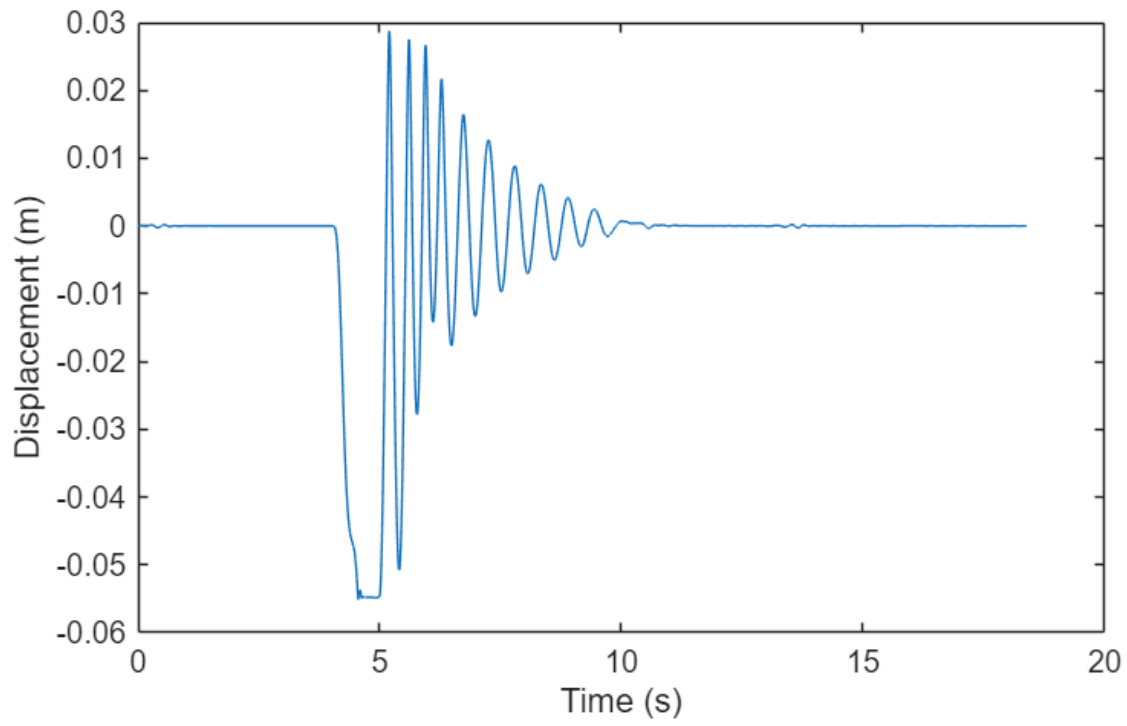


Figure 5. Raw data for supplemental non-linear response.

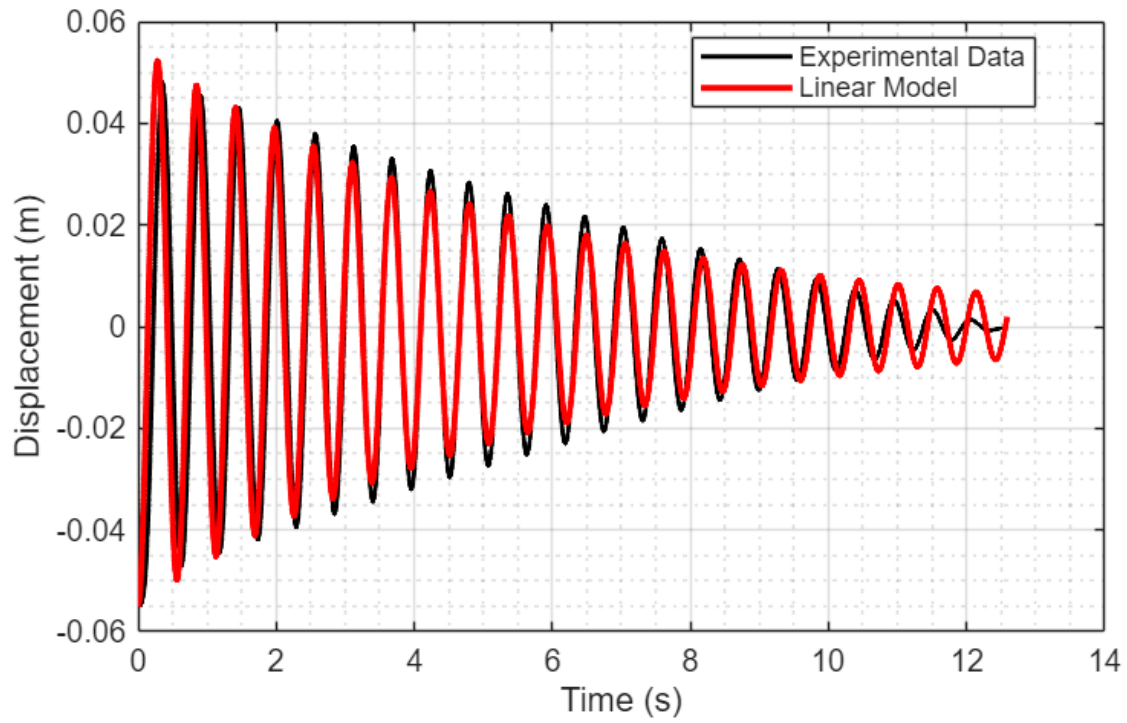


Figure 6. Linear experimental response and analytical solution of zero block configuration.

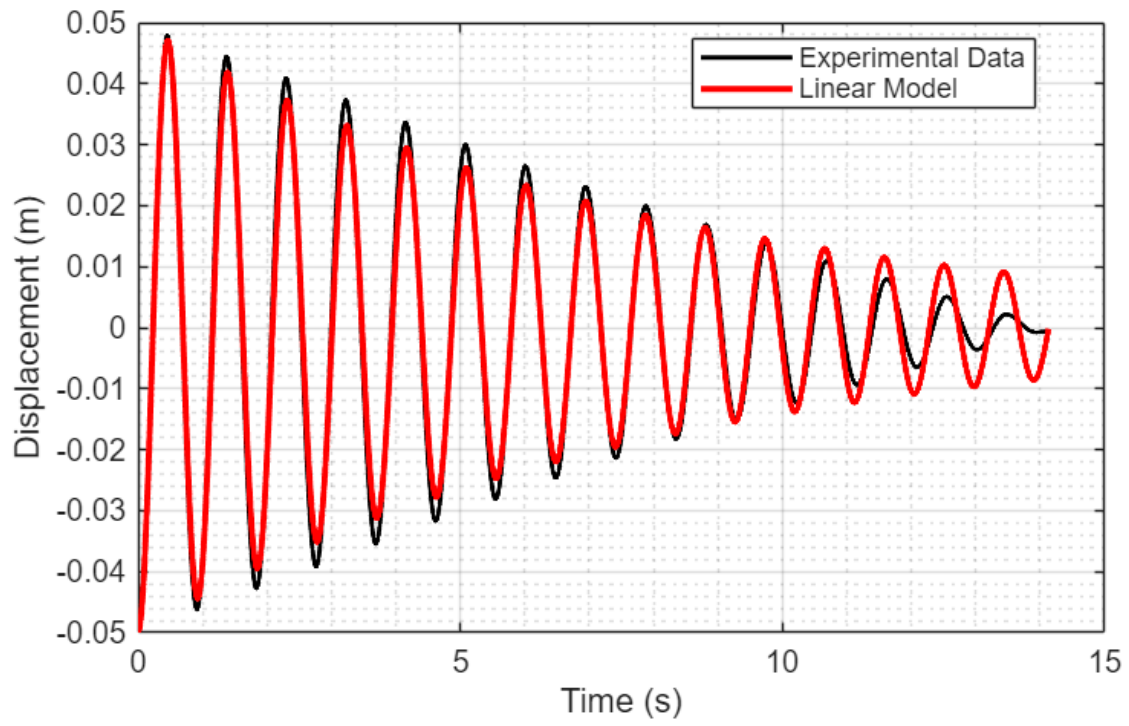


Figure 7. Linear experimental response and analytical solution of four blocks configuration.

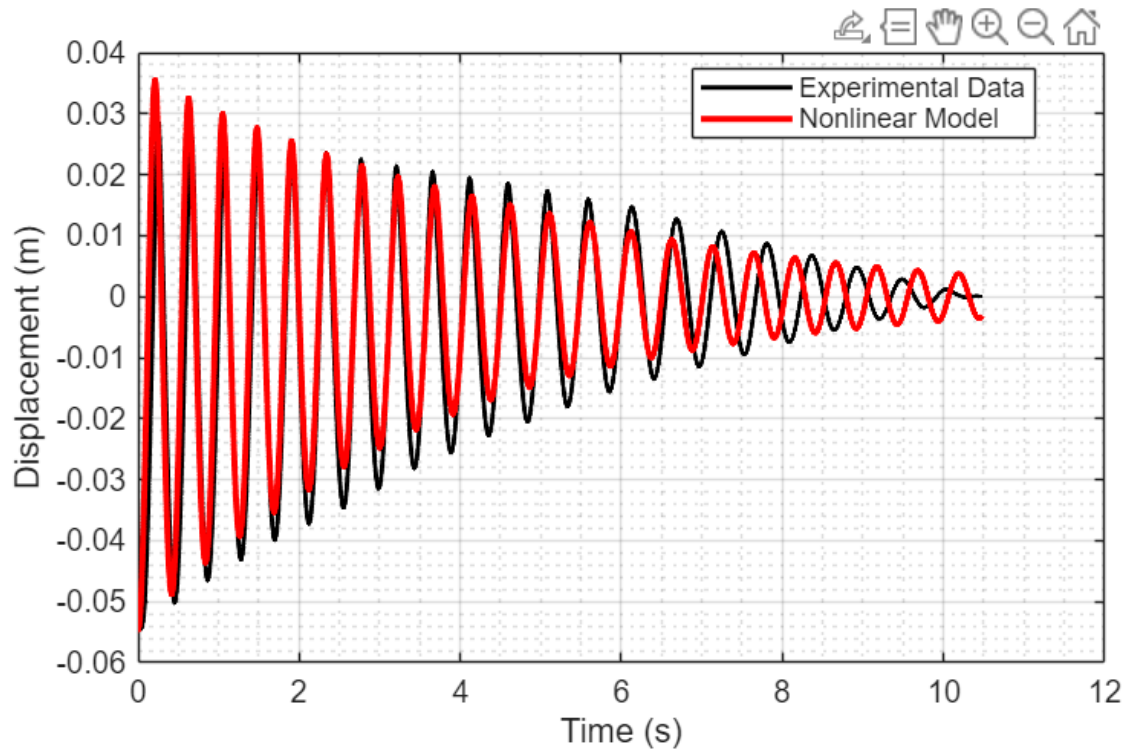


Figure 8. Non-linear experimental response and analytical solution of zero block configuration.

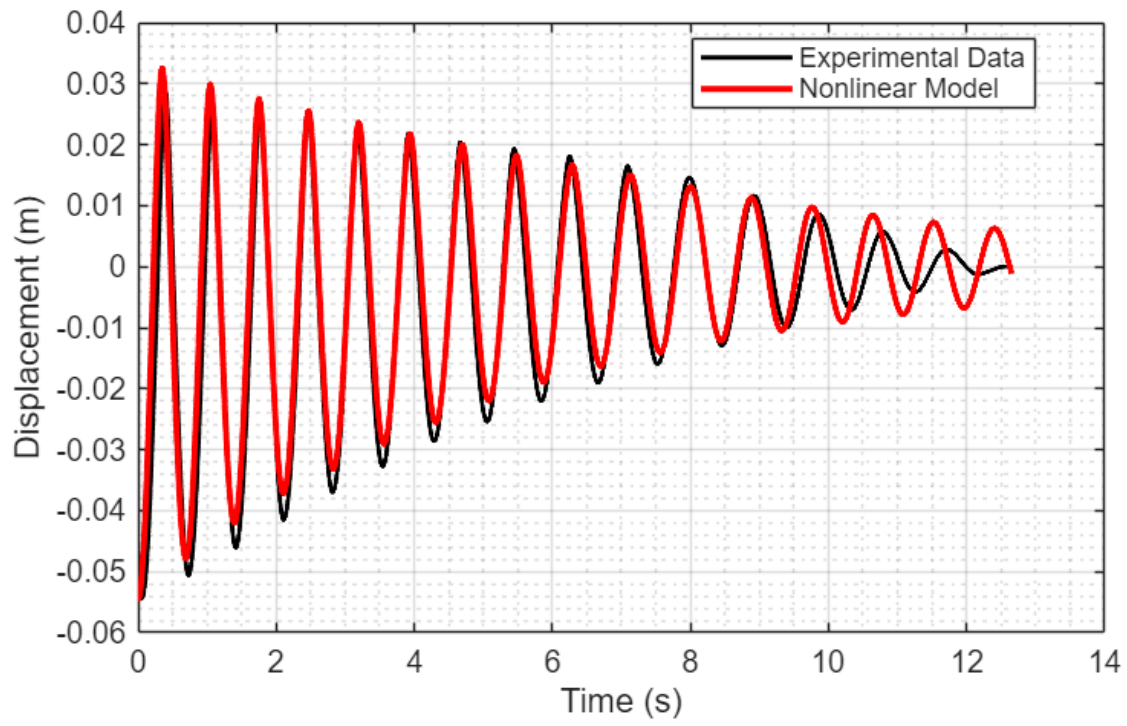


Figure 9. Non-linear experimental response and analytical solution of four blocks configuration.

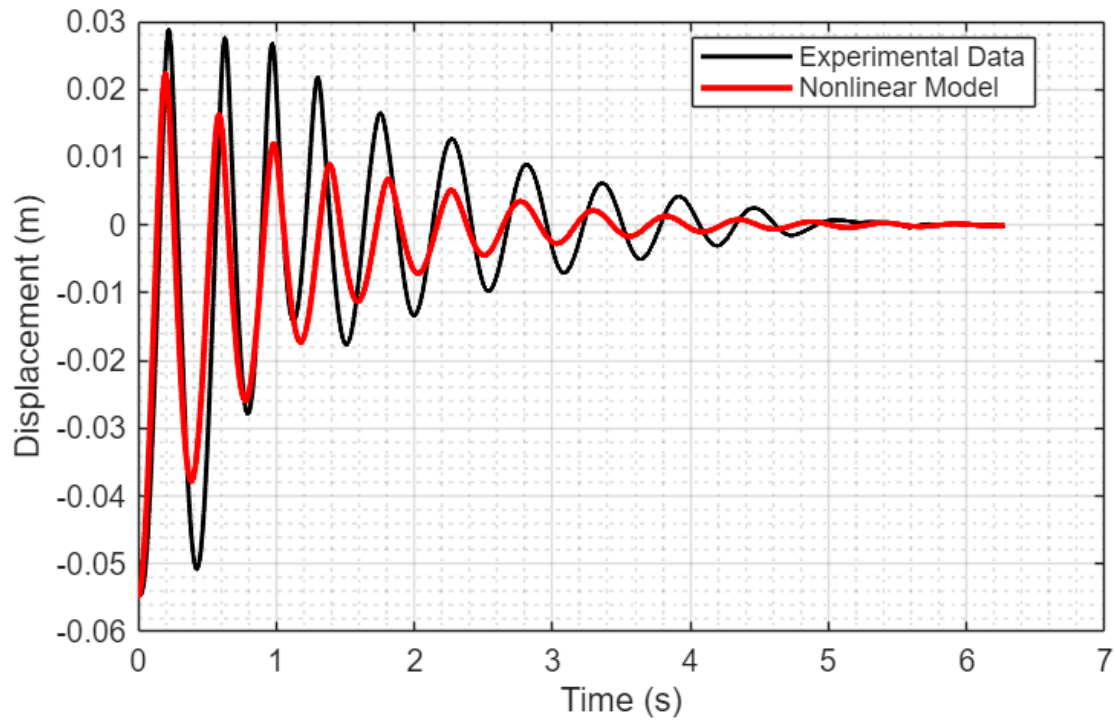


Figure 10. Supplemental non-linear experimental response and analytical solution.